

“Fitness as National Capital: A Simulation Model of Physical Fitness, Health, and Economic Outcomes”

Jagrič, T., Jung, J. M., Došenović Bonča, P., Starc, G., Leskošek, B., Martinko, A., Sorić, M., Jurak, G.

1. Introduction

As populations age, physical inactivity is placing increasing pressure on health systems, labor markets, and public finances. It contributes not only to a greater burden of chronic disease and rising healthcare expenditure but also to lower labor productivity, greater absenteeism, and earlier withdrawal from the labor force (OECD, 2019; WHO, 2022). A substantial literature documents these consequences. Physical inactivity is associated with premature mortality, obesity, cardiovascular disease, type 2 diabetes, and other chronic conditions, whereas regular physical activity and higher functional capacity are associated with lower mortality risk, better health, and improved quality of life (Lee et al., 2012; Guthold et al., 2018; Warburton & Bredin, 2017). These health improvements may also generate broader economic benefits through higher labor-force participation, greater productivity, lower absenteeism, and delayed retirement (Bloom et al., 2014; Suhrcke et al., 2006).

Despite this evidence, the health and economic consequences of physical activity are often examined separately. Epidemiological studies typically focus on morbidity, mortality, obesity, or quality of life, while studies in Economics emphasize healthcare expenditure, productivity, absenteeism, and labor-force participation. As a result, existing research provides limited insight into how improvements in physical fitness accumulate over the life course and jointly affect population health, public expenditure, and economic performance.

A related limitation is that economic and epidemiological models treat physical activity as an exogenous behavioral input. Although participation in physical activity is crucial, it does not directly measure the physiological consequences of that behavior. Individuals reporting similar activity levels may experience different changes in aerobic capacity, strength, mobility, and overall functional ability. Functional capacity may therefore provide a more direct

mechanism through which preventive interventions affect obesity, healthy life years, healthcare expenditure, work ability, productivity, and retirement.

This distinction is particularly important for long-term policy evaluation. Physical activity describes what individuals do, whereas functional capacity captures the accumulated physical condition resulting from activity, aging, health, and other life-course exposures. A framework centered on functional capacity can therefore provide a clearer representation of the pathway from preventive intervention to physiological change and, subsequently, to health and economic outcomes.

Slovenia's SLOfit monitoring system provides a valuable empirical foundation for this approach. SLOfit offers standardized, long-term information on physical and functional capacity and makes it possible to anchor the analysis in observed measures rather than relying exclusively on self-reported activity or aggregate participation rates. These measures can be used to define an empirically plausible functional-capacity scale and to constrain simulated policy effects to ranges supported by observed population data.

This study develops an integrated simulation model linking functional capacity to health, fiscal, labor-market, and broader economic outcomes. The model represents functional capacity using a standardized functional-capacity indicator, or FC score, and traces several interconnected pathways. Improvements in functional capacity affect obesity and healthy life years, which in turn influence healthcare expenditure. The framework also incorporates productivity, absenteeism, early retirement, labor-force participation, and fiscal feedback.

The model is used to compare five preventive-policy scenarios: school-based interventions for children and adolescents, active mobility, prevention among working-age adults, functional aging among older adults, and an integrated life-course strategy. The scenarios differ in their target populations, participation rates, adherence, costs, and expected effects on functional capacity. Their consequences are evaluated using a common set of outcomes, including healthy life years, healthcare savings, incremental cost-effectiveness ratios, returns on investment, benefit-cost ratios, and social net benefits.

This study makes three contributions. First, it develops a unified framework connecting physical capacity to population health, healthcare expenditure, labor-market outcomes, and economic welfare. Second, it uses standardized SLOfit measures to establish an empirically grounded physiological transmission mechanism between preventive interventions and subsequent outcomes. Third, it provides a common policy-evaluation framework through which interventions targeting different stages of the life course can be compared consistently. By integrating these pathways, this study provides a quantitative framework for examining physical capacity as an important population-level health and economic resource. The analysis is intended not to predict the exact consequences of any single intervention, but to support transparent comparison of alternative policy strategies and identify the conditions under which investments in physical capacity may generate long-term health, fiscal, and social benefits.

In its implementation, the model is not merely a conceptual framework but an extensive computational system. The program code contains 35 cells, 2,752 lines of code, 27 functions, and 530 explicit computational assignments or transformations, from which the model equations and auxiliary calculations are derived. At the top level it uses 57 global constants, parameters, and configuration settings, 290 uniquely defined variables, and 7 explicitly estimated regression equations. The empirical database includes 759 country-years in the combined panel, a total of 3,742 input observations across the main indicators, 11 SLOfit anchor points, 6 scenarios (5 of which are public-policy scenarios), and a simulation horizon of 25 years. The results are exported to 27 Excel sheets and 47 graphical displays. The model therefore represents a complex, empirically calibrated simulation system that goes beyond standard partial cost-benefit analysis and enables traceable evaluation of the health, fiscal, and broader economic effects of public-policy scenarios.

2. Literature Review

Evidence on the relationship between physical activity and economic outcomes is extensive but fragmented across several distinct strands. The first strand of the literature examines the epidemiological and health effects of physical inactivity. Lee et al. (2012) estimate that physical inactivity accounts for approximately 6–10% of the global burden of major non-

communicable diseases. Similarly, Guthold et al. (2018) documents that levels of physical inactivity are not improving in many countries, indicating a persistent long-term threat to population health. Warburton and Bredin (2017) further emphasizes the broad and mutually reinforcing health benefits of physical activity, including lowering mortality risk and improved functional capacity.

A second strand of the literature investigates the relationship between population health and economic outcomes. Bloom et al. (2014) show that better population health contributes significantly to economic growth, particularly through labor productivity. Suhrcke et al. (2006) similarly highlights that better health is associated with lower absenteeism and greater labor-force participation. At the firm level, these effects may also translate into reduced costs from sick leave and improved worker performance.

The third strand focuses on the economic evaluation of preventive policies. OECD (2019) finds that many preventive interventions, especially those targeting physical activity and nutrition, can be cost-effective and, in some cases, even cost-saving. WHO (2022) emphasizes the potential of population-wide strategies, including infrastructure investments that promote active mobility, to generate substantial health benefits due to their broad reach. Taken together, these studies suggest that interventions targeting the entire population may produce more favorable cost-benefit ratios than narrowly targeted individual-level programs.

Yet important limitations remain. Existing studies rely on aggregate measures, such as physical-activity prevalence or obesity rates, without explicitly linking physical activity to measurable changes in functional capacity. However, physical activity represents a behavior, whereas functional capacity captures its accumulated physiological consequences and thus provides a more direct mechanism connecting activity to health and economic outcomes. In addition, these transmission channels are often modeled separately, making it difficult to trace the full pathway from preventive interventions to physiological change, health outcomes, healthcare expenditure, and broader economic consequences.

Although recent microsimulation and life-course models have attempted to address this fragmentation, they require extensive longitudinal data that are not often available (Briggs et al., 2016). This study addresses these limitations by using standardized, long-term SLOfit measures

of functional capacity as the central transmission variable linking policy interventions to obesity, healthy life years, healthcare expenditure, productivity, absenteeism, and early retirement. By integrating these relationships within a single framework, the model allows alternative interventions to be evaluated consistently across the life course and from both health and economic perspectives.

3. Methodology

The model is designed as an empirically anchored structural-simulation model that, in annual time steps for the period from 2026 to 2050, estimates the effects of alternative SLOfit-oriented preventive policies on physical capacity, obesity, healthy life years, health expenditure, broader social benefits, cost-effectiveness, and additional effects on the labor market and GDP. The implementation first establishes the data foundation, then estimates or calibrates the key parameters, next run a deterministic simulation by scenario and year, followed by a Monte Carlo uncertainty analysis, and finally the extended labor market/GDP layer. Discounting is carried out using an annual discount rate of $r = 0,03$, and the willingness-to-pay thresholds are set at 10,000, 20,000 and 40,000 EUR per additional healthy life year.

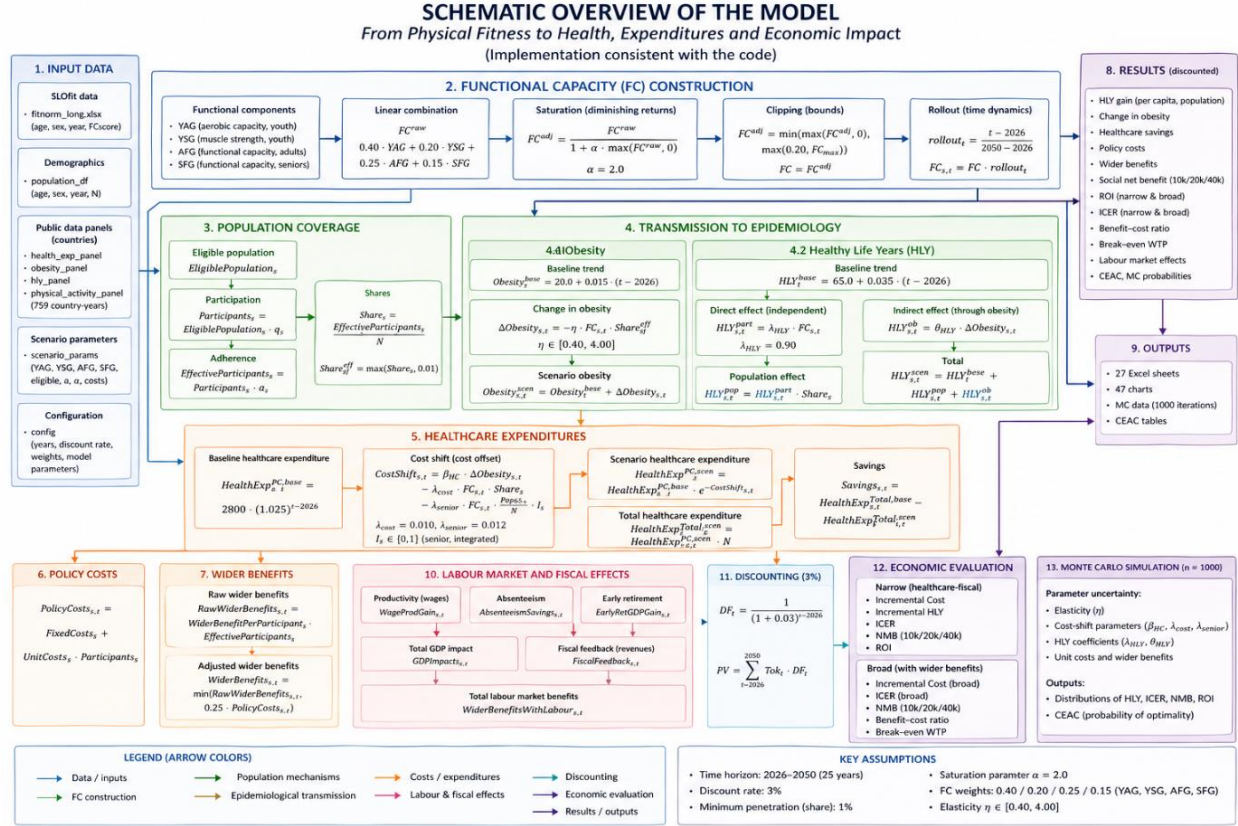


Figure1 : Model Design

The demographic basis of the model is the actual age structure of Slovenia, obtained from Eurostat's demo_pjan database. From the age distribution the total population is calculated as

$$N = \sum_a P_o p_a,$$

together with the target population groups

$$Pop_{6-19} = \sum_{a=6}^{19} P_o p_a, Pop_{20-64} = \sum_{a=20}^{64} P_o p_a, Pop_{65+} = \sum_{a \geq 65} P_o p_a,$$

and

$$Pop_{6+} = \sum_{a \geq 6} P_o p_a.$$

These groups directly define the eligible population for each scenario. The school scenario uses the population Pop_{6-19} , the adult-prevention scenario uses Pop_{20-64} , the senior scenario uses Pop_{65+} , active mobility uses Pop_{6+} , and the integrated scenario uses the entire population N .

The central bridge between SLOfit logic and the model's effects is the latent functional-capacity variable, denoted FC . SLOfit data are not used in the model as a micro-level database, but rather as public anchor data. The model constructs a reconstruction template using publicly grounded anchors, which include the year, the mean fitness percentile, the 10th percentile, the 90th percentile, the obesity rate, and the share of low physical capacity. The fitness percentiles are converted into a standardized functional measure using the equation :

$$FC_{mean,t} = \frac{P_{mean,t} - 50}{50}.$$

The lower and upper percentile edges are converted in the same way:

$$FC_{10,t} = \frac{P_{10,t} - 50}{50}, FC_{90,t} = \frac{P_{90,t} - 50}{50}.$$

From these values, the empirical bounds of functional capacity are determined:

$$FC_{\min} = \min_t FC_{10,t}, FC_{\max} = \max_t FC_{90,t}.$$

The first transmission channel of the model is therefore the channel transmitting SLOfit into functional capacity. This channel does not create a direct health effect but instead defines the measurement scale within which scenario effects can move. This prevents scenarios from generating functional changes outside the empirically observed range of the SLOfit anchor data.

The second transmission channel is the channel transmitting SLOfit into obesity. Using the SLOfit anchor data, the relationship between the obesity rate and the observed functional index is estimated as:

$$Obesity_t = \alpha + \beta_{SLOfit} FC_{mean,t} + \gamma t + \varepsilon_t.$$

Since higher functional capacity is expected to be associated with lower obesity, the estimated coefficient is converted into a positive reduction elasticity:

$$\eta_{SLOfit} = \max(0, -\beta_{SLOfit}).$$

If international physical-activity data are also available, the model additionally estimates the relationship between physical activity and obesity on a European panel:

$$Obesity_{c,t} = \alpha + \beta_{PA}PhysicalActivity_{c,t} + \gamma Age65_{c,t} + \delta t + \mu_c + \varepsilon_{c,t}.$$

Here as well, the coefficient is converted into a positive reduction parameter:

$$\eta_{PA} = \max(0, -\beta_{PA}).$$

The final elasticity of transmission from functional capacity into obesity is determined hierarchically. If both the SLOfit estimate and the international estimate are available, the model uses a weighted combination:

$$\eta = 0,70\eta_{SLOfit} + 0,30\eta_{PA}.$$

If only the SLOfit estimate is available, then

$$\eta = \eta_{SLOfit}.$$

If only the international estimate is available, then

$$\eta = \eta_{PA}.$$

If neither estimate is available, the model uses a fallback value of 1.50. The final value is bounded to the interval

$$0,40 \leq \eta \leq 4,00.$$

This parameter is denoted in the model as FITNESS_TO_OBESITY_ELASTICITY and is key to the epidemiological transmission of functional capacity into the obesity rate.

Before the scenario simulation begins, the model also estimates two European health regression parameters. The first is the effect of obesity on health expenditure. The regression takes the form as following,

$$\ln(HealthExpPC_{c,t}) = \alpha + \beta_{HC}Obesity_{c,t} + \gamma Age65_{c,t} + \delta t + \mu_c + \varepsilon_{c,t}.$$

Estimation is carried out using OLS with HC3 robust standard errors. Country fixed effects are included via $C(country)$. The regression is run if at least 40 complete observations are available. From the regression, the coefficient β_{HC} is used, which is bounded to the interval

$$0,002 \leq \beta_{HC} \leq 0,040.$$

If the regression is not feasible or the estimate is not finite, the model uses a fallback value of 0.018. The standard error of this coefficient is bounded, for the Monte Carlo component, to the interval

$$0,002 \leq SE(\beta_{HC}) \leq 0,020,$$

with a fallback value of 0.006.

The second health regression parameter is the effect of obesity on healthy life years. The regression is

$$HLY_{c,t} = \alpha + \beta_{HLY} Obesity_{c,t} + \gamma Age65_{c,t} + \delta t + \mu_c + \varepsilon_{c,t}.$$

This regression is also estimated using OLS, HC3 robust standard errors, and country fixed effects. The coefficient β_{HLY} is bounded to the interval

$$-0,120 \leq \beta_{HLY} \leq -0,005.$$

If the regression is not feasible, a fallback value of $-0,035$ is used. The standard error is used in the Monte Carlo analysis and is bounded to

$$0,005 \leq SE(\beta_{HLY}) \leq 0,060,$$

with a fallback value of 0.015.

The third component of the model's core consists of the scenario channels of functional capacity. Each scenario is described by four physiological-functional components: improvement in youth aerobic capacity, improvement in youth strength, functional gain among adults, and functional gain among the elderly. The raw functional index of a scenario is a weighted sum of these channels:

$$FC_s^{raw} = 0,40YAG_s + 0,20YSG_s + 0,25AFG_s + 0,15SFG_s.$$

Here YAG_s is youth_aerobic_gain, YSG_s is youth_strength_gain, AFG_s is adult_functional_gain, and SFG_s is senior_functional_gain. This means the model does not treat fitness as a single homogeneous variable, but as a composite of four transmission channels that are activated to different degrees across scenarios.

Since the model does not allow linear summation of effects without a limit, a diminishing-marginal-returns function is applied:

$$FC_s^{adj} = \frac{FC_s^{raw}}{1 + \alpha \max(FC_s^{raw}, 0)}.$$

The diminishing-returns parameter is

$$\alpha = 2,0.$$

The adjusted index is then bounded by the empirical SLOfit range:

$$FC_s^{adj} = \min\left(\max(FC_s^{adj}, 0), \max(0, 20, FC_{\max})\right).$$

The timing of policy implementation is incorporated through a linear rollout factor:

$$rollout_t = \frac{t - 2026}{2050 - 2026}.$$

The annual scenario functional index is therefore

$$FC_{s,t} = FC_s^{adj} \cdot rollout_t.$$

The fourth transmission channel is the coverage and adherence channel. For each scenario, the number of participants is calculated as the product of the eligible population and the target enrollment rate:

$$Participants_s = EligiblePopulation_s \cdot q_s.$$

Effective participants are

$$EffectiveParticipants_s = Participants_s \cdot a_s,$$

where a_s is adherence, i.e. the effectiveness of actual participation. The effective population share is

$$Share_s = \frac{EffectiveParticipants_s}{N}.$$

This share is used when transmitting functional capacity into population-level effects. When transmitting into obesity, the model applies a lower bound of 0.01, so the relevant share is

$$\max(Share_s, 0.01).$$

The fifth transmission channel is the epidemiological channel from functional capacity into obesity. The change in obesity, in percentage points, is

$$\Delta Obesity_{s,t} = -\eta \cdot FC_{s,t} \cdot \max(Share_s, 0.01).$$

The baseline obesity rate is an exogenous linear trend:

$$Obesity_t^{base} = 20.0 + 0.015(t - 2026).$$

The scenario obesity rate is

$$Obesity_{s,t}^{scen} = Obesity_t^{base} + \Delta Obesity_{s,t}.$$

The sixth transmission channel is the direct functional channel into healthy life years. The baseline HLY value follows a linear trend:

$$HLY_t^{base} = 65.0 + 0.035(t - 2026).$$

The participant HLY bonus is

$$HLY_{s,t}^{participant} = \lambda_{HLY} FC_{s,t},$$

where

$$\lambda_{HLY} = 0.90.$$

Since this effect applies to participants, at the population level it is multiplied by the effective population share:

$$HLY_{s,t}^{population} = HLY_{s,t}^{participant} \cdot Share_s.$$

The seventh transmission channel is the indirect channel from obesity into HLY. Since the change in obesity affects healthy life years via the regression coefficient β_{HLY} , this effect is

$$HLY_{s,t}^{obesity} = \beta_{HLY} \Delta Obesity_{s,t}.$$

Total scenario HLY is the sum of the baseline value, the direct population-level functional effect, and the indirect effect via obesity:

$$HLY_{s,t}^{scen} = HLY_t^{base} + HLY_{s,t}^{population} + HLY_{s,t}^{obesity}.$$

The annual HLY gain relative to baseline is

$$\Delta HLY_{s,t} = HLY_{s,t}^{scen} - HLY_t^{base}.$$

The eighth transmission channel is the channel from obesity and functional capacity into health expenditure. Baseline health expenditure per capita grows at a rate of 2.5% annually:

$$HealthExpPC_t^{base} = 2800(1 + 0,025)^{t-2026}.$$

Total baseline health expenditure is

$$HealthExpTotal_t^{base} = HealthExpPC_t^{base} \cdot N.$$

The scenario cost shift is a logarithmic shift in expenditure and has three components. The first is the effect of obesity, the second is the direct functional cost effect, and the third is the senior functional cost effect:

$$CostShift_{s,t} = \beta_{HC} \Delta Obesity_{s,t} - \lambda_{cost} FC_{s,t} Share_s - \lambda_{senior} FC_{s,t} \frac{Pop_{65+}}{N} I_s.$$

Here

$$\lambda_{cost} = 0,010, \lambda_{senior} = 0,012.$$

The indicator I_s equals 1 for the senior_functional_ageing and integrated_life_course scenarios, and 0 otherwise. Scenario health expenditure per capita is

$$HealthExpPC_{s,t}^{scen} = HealthExpPC_t^{base} \exp(CostShift_{s,t}).$$

Total scenario health expenditure is

$$HealthExpTotal_{s,t}^{scen} = HealthExpPC_{s,t}^{scen} N.$$

Health savings are

$$Savings_{s,t} = HealthExpTotal_t^{base} - HealthExpTotal_{s,t}^{scen}.$$

The savings rate is

$$SavingsRate_{s,t} = \frac{Savings_{s,t}}{HealthExpTotal_t^{base}}.$$

The ninth channel is the cost-of-policy channel. The annual cost of the policy is

$$PolicyCost_{s,t} = FixedCost_s + UnitCost_s \cdot Participants_s.$$

The raw wider benefit is

$$\begin{aligned} & RawWiderBenefit_{s,t} \\ &= WiderBenefitPerEffectiveParticipant_s \cdot EffectiveParticipants_s. \end{aligned}$$

To prevent the model from overestimating wider benefits, the final wider benefit is capped at 25% of the annual policy cost:

$$WiderBenefit_{s,t} = \min(RawWiderBenefit_{s,t}, 0.25 PolicyCost_{s,t}).$$

The narrow total cost is

$$TotalCost_{s,t} = HealthExpTotal_{s,t}^{scen} + PolicyCost_{s,t}.$$

The broad total cost is

$$TotalCostBroad_{s,t} = TotalCost_{s,t} - WiderBenefit_{s,t}.$$

The annual net effect in the narrow health-budget perspective is

$$AnnualNetSavings_{s,t} = Savings_{s,t} - PolicyCost_{s,t}.$$

The annual broad net effect is

$$AnnualBroadNetBenefit_{s,t} = Savings_{s,t} + WiderBenefit_{s,t} - PolicyCost_{s,t}.$$

The tenth channel is the economic-discounting channel. The discount factor is

$$DF_t = \frac{1}{(1 + 0,03)^{t-2026}}.$$

The discounted present value of any arbitrary flow $X_{s,t}$ is

$$PV(X_s) = \sum_t X_{s,t} DF_t.$$

On this basis the discounted policy costs, health savings, raw wider benefits, capped wider benefits, and HLY effects are calculated. The narrow incremental cost is

$$IncrementalCost_s^{narrow} = PV(PolicyCost_s) - PV(Savings_s).$$

The broad incremental cost is

$$IncrementalCost_s^{broad} = PV(PolicyCost_s) - PV(Savings_s) - PV(WiderBenefit_s).$$

The discounted population-level HLY effect is

$$PV(\Delta HLY_s^{pop}) = N \sum_t \Delta HLY_{s,t} DF_t.$$

The narrow ICER is

$$ICER_s^{narrow} = \frac{IncrementalCost_s^{narrow}}{PV(\Delta HLY_s^{pop})}.$$

The broad ICER is

$$ICER_s^{broad} = \frac{IncrementalCost_s^{broad}}{PV(\Delta HLY_s^{pop})}.$$

The narrow ROI is

$$ROI_s^{narrow} = \frac{PV(Savings_s)}{PV(PolicyCost_s)}.$$

The broad ROI is

$$ROI_s^{broad} = \frac{PV(Savings_s) + PV(WiderBenefit_s)}{PV(PolicyCost_s)}.$$

The net monetary benefit at a willingness-to-pay threshold WTP is

$$NMB_s(WTP) = WTP \cdot PV(\Delta HLY_s^{pop}) - IncrementalCost_s^{narrow}.$$

In the model this is computed for $WTP = 10.000, 20.000$ and 40.000 . The benefit-cost ratio at the 20,000 EUR threshold is

$$BCR_s(20k) = \frac{20000 \cdot PV(\Delta HLY_s^{pop}) + PV(Savings_s) + PV(WiderBenefit_s)}{PV(PolicyCost_s)}.$$

The break-even willingness-to-pay threshold is

$$BreakEvenWTP_s = \frac{IncrementalCost_s^{narrow}}{PV(\Delta HLY_s^{pop})}.$$

The eleventh channel is the stochastic uncertainty channel. The model runs 5000 Monte Carlo repetitions. In each repetition the health-cost semi-elasticity is drawn as

$$\beta_{HC}^{MC} = \max(0,0005, \mathcal{N}(\beta_{HC}, SE_{\beta_{HC}})).$$

The obesity coefficient for HLY is drawn as

$$\beta_{HLY}^{MC} = \min(-0,001, \mathcal{N}(\beta_{HLY}, SE_{\beta_{HLY}})).$$

The elasticity of functional capacity into obesity is

$$\eta^{MC} = \max(0,05, \mathcal{N}(\eta, 0,30\eta)).$$

The direct HLY parameter is

$$\lambda_{HLY}^{MC} = \max(0,05, \mathcal{N}(\lambda_{HLY}, 0,25\lambda_{HLY})).$$

The functional cost parameter is

$$\lambda_{cost}^{MC} = \max(0, \mathcal{N}(\lambda_{cost}, 0,40\lambda_{cost})).$$

In each repetition the scenario may fail with probability

$$p_{failure} = 0,10,$$

or produce a negative (backfire) effect with probability

$$p_{backfire} = 0,035.$$

If failure occurs, the effect multiplier is

$$m = 0.$$

If backfire occurs, then

$$m = -U(0,05,0,20).$$

If the scenario succeeds, then

$$m \sim LogNormal(0,0,30).$$

In addition, multipliers for coverage, adherence, cost, and wider benefits are drawn:

$$m_{coverage} \sim U(0,60,1,05), m_{adherence} \sim U(0,65,1,10), m_{cost} \sim LogNormal(0,0,15), m_{wider} \sim LogNormal(0,0,25).$$

In each Monte Carlo repetition the entire deterministic calculation for the scenario and year is then repeated, but with stochastically varied parameters. From the results the mean, 5th percentile, median, 95th percentile, probability of a positive HLY effect, failure rate, backfire rate, and percentile distribution of the ICER are calculated. The probability of a positive effect is

$$P(HLY_s > 0) = \frac{1}{R} \sum_{r=1}^R I(HLY_{s,r}^{MC} > 0),$$

where $R = 5000$.

The twelfth channel is the CEAC channel. The model does not merely compute the probability that a scenario is cost-effective relative to a null alternative, but rather the probability that it is optimal among all scenarios. For each repetition and each threshold WTP , the scenario with the highest net monetary benefit is determined:

$$s_r^*(WTP) = \operatorname{argmax}_s NMB_{s,r}(WTP).$$

The probability of optimality is

$$P_s^{optimal}(WTP) = \frac{1}{R} \sum_{r=1}^R I(s_r^*(WTP) = s).$$

The thirteenth channel is the labor market/GDP channel. The model first attempts to estimate three additional regression relationships. For absenteeism the following is used:

$$HealthAbsence_{c,t} = \alpha + \beta_{abs}PhysicalActivity_{c,t} + \gamma Obesity_{c,t} + \delta Age65_{c,t} + \mu_c + \varepsilon_{c,t}.$$

Since more physical activity is expected to reduce absenteeism, the coefficient is constrained such that

$$\beta_{abs} \leq 0.$$

If the regression is not feasible, a fallback value of $-0,015$ is used. From this the number of absence days avoided per unit of the functional index is calculated:

$$DaysAvoidedPerFC = clip\left(\frac{|\beta_{abs}| \cdot 14}{10}, 0,05, 1,50\right).$$

For older-worker employment the following is estimated:

$$\begin{aligned} &OlderWorkerEmployment_{c,t} \\ &= \alpha + \beta_{old}PhysicalActivity_{c,t} + \gamma Obesity_{c,t} + \delta Age65_{c,t} + \mu_c + \varepsilon_{c,t}. \end{aligned}$$

The coefficient is constrained to a positive sign:

$$\beta_{old} \geq 0.$$

If the regression is not feasible, a fallback value of 0.020 is used. The parameter for avoided early retirement is

$$EarlyRetirementRatePerFC = clip\left(\frac{\beta_{old}}{100}, 0,0001, 0,0100\right).$$

For macro productivity the following is estimated:

$$GDPPC_{c,t} = \alpha + \beta_{gdp}PhysicalActivity_{c,t} + \gamma Obesity_{c,t} + \delta Age65_{c,t} + \mu_c + \varepsilon_{c,t}.$$

Here too the coefficient is constrained to a positive sign:

$$\beta_{gap} \geq 0.$$

If the regression is not feasible, a fallback value of 35.0 is used. The productivity parameter relative to GDP per worker is

$$GDPProductivityGainPerFC = clip\left(\frac{\beta_{gap}}{GDPPerWorker}, 0,0001,0,0200\right).$$

Since the model uses a conservative wage-based version of productivity, the following holds:

$$\begin{aligned} WageProductivityGainPerFC \\ = clip(0,50 \cdot GDPProductivityGainPerFC, 0,0001,0,0100). \end{aligned}$$

Here the model uses

$$AverageGrossAnnualWage = 32000, GDPPerWorker = 62000, WorkingDays = 220.$$

The cost of a workday is

$$WageCostPerWorkday = \frac{32000}{220}.$$

The fiscal feedback parameter is

$$\tau = 0,35.$$

The fourteenth channel is the split of HLY between working age and non-working age. The annual population-level HLY effect is

$$HLYGain_{s,t}^{pop} = \Delta HLY_{s,t} N.$$

The working-age HLY effect is

$$HLYGain_{s,t}^{work} = w_s HLYGain_{s,t}^{pop}.$$

The non-working-age effect is

$$HLYGain_{s,t}^{nonwork} = HLYGain_{s,t}^{pop} - HLYGain_{s,t}^{work}.$$

The weights w_s are scenario-specific: 0.35 for school_push, 0.70 for active_mobility, 0.90 for adult_prevention, 0.10 for senior_functional_ageing, and 0.60 for integrated_life_course.

The fifteenth channel is the conversion of functional capacity into productivity, absenteeism, and early retirement. Effective working-age participants are

$$EffectiveParticipants_s^{work} = w_s EffectiveParticipants_s.$$

Effective older workers are

$$EffectiveParticipants_s^{older} = v_s EffectiveParticipants_s,$$

where the weights v_s equal 0.00 for school_push, 0.15 for active_mobility, 0.20 for adult_prevention, 0.10 for senior_functional_ageing, and 0.15 for integrated_life_course.

The wage productivity gain is

$$\begin{aligned} & WageProductivityGain_{s,t} \\ &= EffectiveParticipants_s^{work} \cdot AverageWage \cdot WageProductivityGainPerFC \cdot FC_{s,t}. \end{aligned}$$

Avoided sick-leave days are

$$SickLeaveDaysAvoided_{s,t} = EffectiveParticipants_s^{work} \cdot DaysAvoidedPerFC \cdot FC_{s,t}.$$

The monetized savings from absenteeism are

$$AbsenteeismSavings_{s,t} = SickLeaveDaysAvoided_{s,t} \cdot WageCostPerWorkday.$$

Avoided early retirement is

$$\begin{aligned} & EarlyRetirementAvoided_{s,t} \\ &= EffectiveParticipants_s^{older} \cdot EarlyRetirementRatePerFC \cdot FC_{s,t}. \end{aligned}$$

The effect of this channel on GDP is

$$EarlyRetirementGDPGain_{s,t} = EarlyRetirementAvoided_{s,t} \cdot GDPPerWorker.$$

The total GDP effect is

$$\begin{aligned}
& GDPImpact_{s,t} \\
& = WageProductivityGain_{s,t} + AbsenteeismSavings_{s,t} + EarlyRetirementGDPGain_{s,t}.
\end{aligned}$$

The fiscal feedback effect is

$$FiscalFeedback_{s,t} = \tau GDPImpact_{s,t}.$$

The wider benefit including the labor-market layer is

$$\begin{aligned}
& WiderBenefit_{s,t}^{labour} \\
& = WiderBenefit_{s,t} + WageProductivityGain_{s,t} + AbsenteeismSavings_{s,t} \\
& \quad + EarlyRetirementGDPGain_{s,t}.
\end{aligned}$$

The broad cost including the labor-market layer is

$$TotalCostBroad_{s,t}^{labour} = TotalCost_{s,t} - WiderBenefit_{s,t}^{labour}.$$

The annual fiscal net effect is

$$AnnualFiscalNetBenefit_{s,t} = Savings_{s,t} + FiscalFeedback_{s,t} - PolicyCost_{s,t}.$$

The annual social net benefit including the labor-market layer, at threshold WTP , is

$$\begin{aligned}
& AnnualSocialNetBenefit_{s,t}^{labour}(WTP) \\
& = WTP \cdot HLYGain_{s,t}^{pop} + Savings_{s,t} + WiderBenefit_{s,t}^{labour} - PolicyCost_{s,t}.
\end{aligned}$$

The discounted summary then calculates the fiscal incremental cost after the feedback effect

$$FiscalIncrementalCost_s = PV(PolicyCost_s) - PV(Savings_s) - PV(FiscalFeedback_s),$$

the broad incremental cost

$$\begin{aligned}
& BroadIncrementalCost_s^{labour} \\
& = PV(PolicyCost_s) - PV(Savings_s) - PV(WiderBenefit_s^{labour}),
\end{aligned}$$

the labor-market-extended ICER

$$ICER_s^{labour} = \frac{BroadIncrementalCost_s^{labour}}{PV(\Delta HLY_s^{pop})},$$

the labor-market-extended benefit-cost ratio at 20,000 EUR

$$BCR_s^{labour}(20k) = \frac{20000 \cdot PV(\Delta HLY_s^{pop}) + PV(Savings_s) + PV(WiderBenefit_s^{labour})}{PV(PolicyCost_s)},$$

the break-even WTP with this layer

$$BreakEvenWTP_s^{labour} = \frac{BroadIncrementalCost_s^{labour}}{PV(\Delta HLY_s^{pop})},$$

and the social net benefit with this layer

$$SocialNetBenefit_s^{labour}(WTP) = WTP \cdot PV(\Delta HLY_s^{pop}) - BroadIncrementalCost_s^{labour}.$$

The model as a whole therefore follows a linear transmission structure. The SLOfit anchor data determine the functional-capacity scale and the elasticity of transmission into obesity and scenarios activate four functional channels. functional capacity is transmitted into annual effects via the rollout mechanism; these effects are converted into obesity, HLY, health expenditure, policy costs, and wider benefits; results are discounted and evaluated using ICER, NMB, ROI, BCR, and CEAC; uncertainty is then checked via Monte Carlo simulation; and finally part of the HLY effects is split between working age and non-working age and transmitted into productivity, absenteeism, early retirement, GDP, and the fiscal feedback effect.

In the simulation performed, the key parameters were determined using a combination of regression estimation, empirical anchoring to SLOfit data, bounding of estimates to predetermined realistic intervals, and fallback or conservative priors. The regression estimate of the relationship between obesity and health expenditure was based on the Eurostat panel, where the following equation was estimated, $\ln(HealthExpPC_{c,t}) = \alpha + \beta_{HC}Obesity_{c,t} + \gamma Age65_{c,t} + \delta t + \mu_c + \varepsilon_{c,t}$. The raw coefficient estimate was $-0,0055$, with a standard error of $0,0085$. Since the expected and model-permitted sign of this parameter is positive as higher obesity typically increases health expenditure, the final parameter value was set by bounding it to the lower limit of the allowed interval. The value entering the simulation is therefore $\beta_{HC} = 0,002$.

Similarly, the regression for the relationship between obesity and healthy life years was estimated in the following equation, $HLY_{c,t} = \alpha + \beta_{HLY}Obesity_{c,t} + \gamma Age65_{c,t} + \delta t + \mu_c + \varepsilon_{c,t}$. The estimate was 0,0352, with a standard error of 0,2034. Since the model expects a negative sign as higher obesity reduces healthy life years, the estimate was bounded to the upper limit of the negative interval allowed. The value entering the simulation is therefore $\beta_{HLY} = -0,005$. This means the model applies a very conservative assumption about the effect of obesity on HLY, since it did not use a stronger negative effect but rather the mildest negative value permitted by the calibration interval.

The elasticity of transmission from functional capacity into obesity was primarily determined on the basis of SLOfit public anchor data. These data include reconstructed public aggregates for the 11–14 age group for the period 1990–2023, namely the mean fitness percentile, the 10th and 90th percentiles, the obesity rate, and the share of children with low physical capacity. The percentile data were converted into a standardized FC score. The estimated relationship between obesity and the functional index produced a raw estimate of $-26,6877$ with a standard error of 3,9964. Since such a value would produce an excessively strong transmission from fitness into obesity, it was bounded, in accordance with the model, to the upper limit of the allowed interval. The final elasticity value entering the simulation is therefore $\eta = 4,0$. The empirical range of the functional index, calculated from the SLOfit percentile anchors, runs from $-0,42$ to $0,44$, so scenario functional effects cannot increase without limit beyond the observed SLOfit range.

The model also uses two conservative structural priors. The direct effect of functional capacity on healthy life years is set at $\lambda_{HLY} = 0,9$, meaning that a full unit of the functional index at the participant level contributes 0.9 healthy life years before transmission to the population level. The functional semi-elasticity of health cost risk is set at $\lambda_{cost} = 0,01$. This parameter captures the direct preventive effect of functional capacity on health costs that is not mediated solely through obesity.

The additional labor market/GDP layer was estimated in the simulation on the basis of a regression panel with 61 observations. The regression model for absenteeism produced a final value of $DaysAvoidedPerFC = 0,05$, with $R^2 = 0,963$. This parameter represents the estimated reduction in sick-leave days per effective participant at a full unit of the functional

index. The regression model for older-worker employment, i.e. avoided early retirement, produced a value of $EarlyRetirementRatePerFC = 0,0001$, with $R^2 = 0,969$. The regression model for productivity, i.e. GDP, produced a value of $GDPProductivityGainPerFC = 0,0001$, with $R^2 = 0,997$. Since the model uses a conservative wage-based interpretation of productivity, the parameter $WageProductivityGainPerFC = 0,0001$ is also derived from this. The values are therefore very cautious and do not generate a strong macroeconomic multiplier.

To monetize the labor-market effects, additional reference parameters are used. The average annual gross labor cost is set at 32,000 EUR, GDP per worker at 62,000 EUR, and the fiscal feedback parameter at 0.35. In this way productivity, absenteeism, and avoided early retirement are converted into monetary flows, and the fiscal feedback effect is calculated as the share of the additional GDP or wage effect that flows back into public finances via taxes and contributions.

The coverage of the regression panel shows that the data were unbalanced. For the obesity rate, 179 observations were available; for health expenditure, 409; for HLY, 614; for the share of the elderly, 2,477; and for physical activity, 63 observations. Physical activity was therefore a relatively limited source, which explains why SLOfit anchoring plays the central role in determining the elasticity of fitness on obesity. In the panel, data were available for approximately 44–55 countries or territorial units per year, but with varying availability across indicators. This justifies the use of robust regressions, the bounding of estimates to realistic intervals, and fallback rules that prevent a single unstable regression estimate from producing unrealistic simulation results.

4. Data

This section describes the data used to construct the model, the sources used, the actual scope of available data, and the operational procedure for obtaining, cleaning, and storing data as implemented in the model. The model uses several interlinked data sets, with each indicator explicitly tied to a specific database, URL source, and role within the model.

Demographic structure data come from Eurostat's *demo_pjan* database, which is used to reconstruct the age pyramid and to calculate the share of the population aged 65 or over. This variable enters as a control variable in all key regression models.

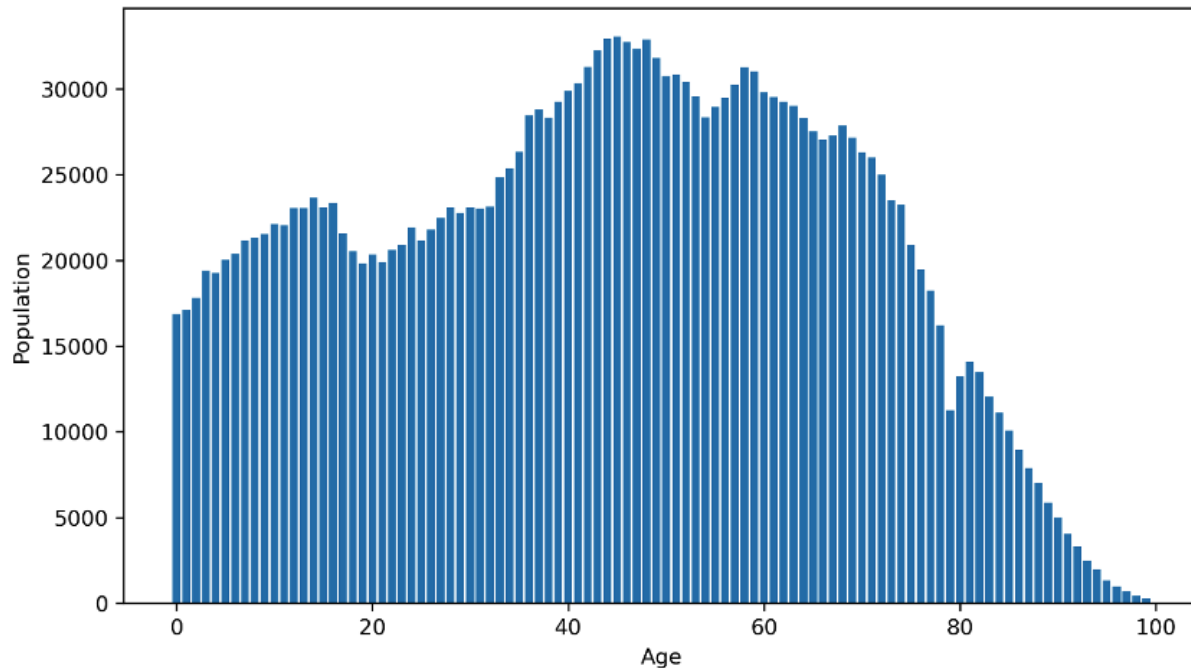


Figure 2 : Population Pyramid of Slovenia used in the production model

Data on the prevalence of obesity are obtained from Eurostat indicator *sdg_02_10* or a compatible body-mass indicator. In the model's actual database, 179 observations are available for this indicator, which sets limits on panel coverage. Data on health expenditure per capita come from the *tps00207* database, for which 409 observations are available. This variable is used in logarithmic form in the health-expenditure regression. Data on healthy life years (HLY) are obtained from the *hlth_hlye* database or a compatible indicator, with 614 observations available in the database. Physical-activity data from the EHIS surveys (*hlth_ehis_pe2e* and related indicators) are available for only 63 observations, and therefore have the status of an optional empirical anchor in the model rather than a central source for parameter identification.

The structure of data availability by year and country is unbalanced, which is explicitly accounted for in the model. Over the period 2010–2024, the number of countries in the panel ranges between 44 and 55 per year, but not all indicators are available simultaneously. For example, in 2014, obesity data are available for 32 countries, health expenditure for 36 countries, HLY for 33 countries, age structure for 47 countries, and physical activity for 32 countries. In

many years, obesity or physical-activity data are not available at all, meaning that the regression models operate on subsets of the panel. This feature of the data directly leads to the use of robust estimates, parameter bounding, and fallback mechanisms in subsequent steps of the model.

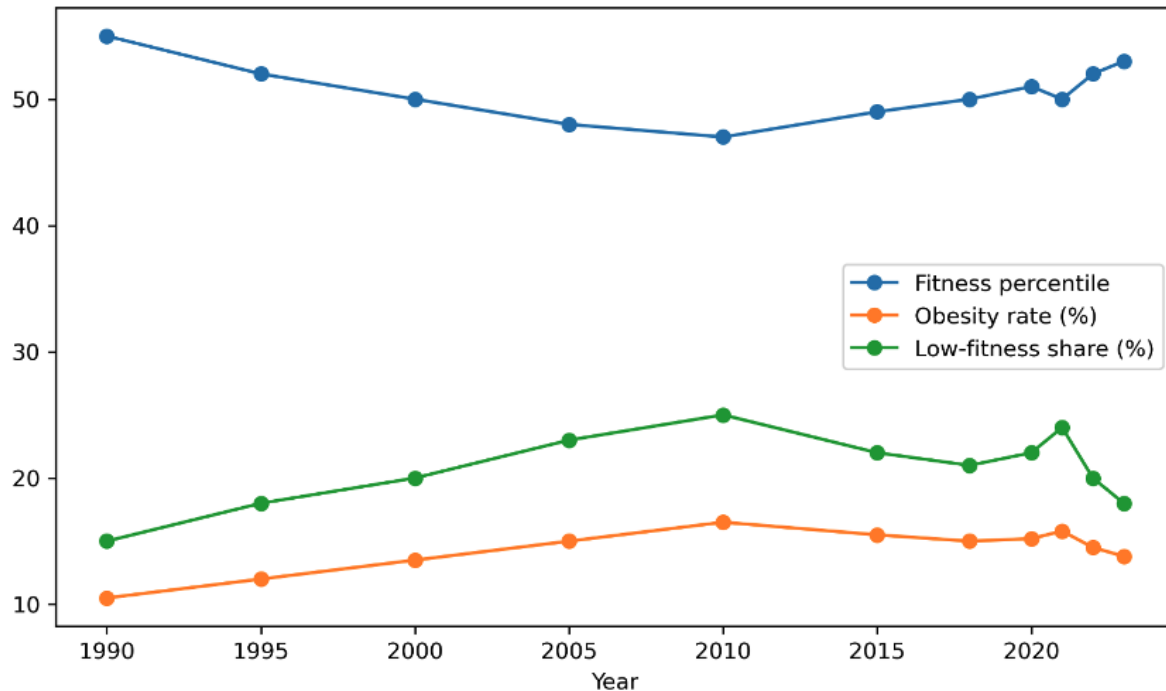


Figure 3 : Public SLOfit anchors used in the transmission layer

A special data layer is represented by the SLOfit public data, which are used as an empirical anchor for the functional index. In the model, these data are organized as a time series for the 11–14 age group for the period 1990–2023 and include the mean fitness percentile, the 10th and 90th percentiles, the obesity rate, and the share of the population with low physical capacity. In addition, the data structure already includes a computed functional index (fcscore) and its percentile edges (p10 and p90). These data carry the status “reconstructed_from_public_reports,” meaning they were reconstructed from SLOfit’s public sources (annual reports and web tools) and then standardized into a single time series. This data layer is used to determine the empirical bounds of the functional index, where the minimum and maximum values are set at $-0,42$ and $0,44$, and for calibrating the elasticity of transmission between functional capacity and obesity.

An additional data layer consists of regression results, which are already stored in the model as parameters together with the number of observations and the coefficient of

determination. These results include models for absenteeism, early retirement, and productivity, with each model based on 61 observations. The coefficients of determination are high: $R^2 = 0,963$ for absenteeism, $R^2 = 0,969$ for early retirement, and $R^2 = 0,997$ for productivity. These results are used directly in the simulation without further re-estimation, meaning the model uses previously estimated parameters as input constants.

Data sources in the model are explicitly linked to URL addresses, which enables reproduction of the results. Eurostat data are obtained via the Data Browser interface, with direct links to individual tables. The WHO's *Health for All* database is used as a fallback source for absenteeism when Eurostat data are not available. OECD productivity data are used as a reference value for GDP per worker. National SURS data serve as an additional validation and contextual benchmark for Slovenia. SLOfit data are obtained from the public web sources of the SLOfit organization. Literature-based priors from the OECD, WHO, and the scientific literature are not obtained automatically but are defined in the model as documented parameters.

The data-acquisition procedure is designed in the model as a modular download by individual indicator. Each indicator is defined by a database identifier, a URL address, and parsing logic. Data are downloaded in tabular form, standardized by country and year, and merged into a panel structure. Internal rules are used for this purpose to harmonize country names, remove missing or clearly inconsistent values, and limit the time span to the period where coverage is sufficient for regression analysis.

After downloading, the data are stored in a structured form that allows direct use in the model. In the model this is implemented as a standardized data frame, where the key dimensions are country, year, and indicator. In addition, individual data layers (e.g., SLOfit, regression results, literature priors) are stored separately, which allows transparent traceability of each parameter. Each parameter in the model has a “source” attribute, which specifies its origin (regression, SLOfit public, Eurostat, literature, or conservative prior), and, where applicable, also the raw estimate and standard error.

This data architecture allows the model to function robustly even under conditions of incomplete data coverage. The combination of direct data sources, empirical anchors, and

predetermined priors ensures that every step of the simulation is based on an explicitly defined input, with each parameter's origin, estimation method, and role in the model fully traceable.

5. Scenario

In this chapter, the scenarios are designed and implemented strictly in accordance with the core of the model from Chapter 3 and with the actual parameters used in the simulation. Scenarios are not given as exogenous shocks but as structured combinations of targeted interventions that, through the functional-capacity (FC) mechanism, generate effects on obesity, healthy life years (HLY), health expenditure, and broader economic indicators. Each scenario is defined by a clearly specified target population, participation rate, adherence rate, and cost structure, with effects introduced gradually over time through a deterministic rollout function.

The scenarios used in the model are: (i) *school_push*, (ii) *active_mobility*, (iii) *adult_prevention*, (iv) *senior_functional_ageing*, and (v) *integrated_life_course*. All scenarios are based on the same total population $N = 2,130,462$, but differ in the size of the target population, the intensity of enrollment, and the expected change in functional capacity.

The *school_push* scenario targets the population of children and adolescents, with an eligible population size of 306,624. Participation amounts to 214,636.8 individuals, and effective participation (accounting for adherence) is 171,709.44, corresponding to an effective population share of $Share = 0.0806$. For this scenario, an initial functional effect of $FC = 0.069$ is used, which is then multiplied over time by the rollout factor. The annual policy cost amounts to 20,731,840 EUR. This scenario represents a high-intensity intervention with relatively limited population reach, meaning individual effects are relatively large, but the aggregate impact is limited by the size of the target group.

The *active_mobility* scenario covers a broader population (2,019,984), with substantially higher participation (706,994.4) and effective participation of 459,546.36, corresponding to a share of $Share = 0.2157$. The initial FC effect used is 0.047, lower than in the *school_push* scenario, but because of the greater reach, a larger aggregate effect is generated. The annual cost amounts to 29,674,860 EUR. This scenario reflects environmental and infrastructural measures,

where the effect per individual is moderate, but the impact on the population is large due to high penetration.

The *adult_prevention* scenario targets the adult population, with a size of 1,241,948, participation of 521,618.16, and effective participation of 323,403.26, corresponding to a share of $Share = 0.1518$. The functional effect is $FC = 0.031$, i.e. relatively low, reflecting the fact that behavior change is harder to achieve among adults. The annual cost amounts to 56,337,543.6 EUR, the highest cost per participant among the individual segment scenarios. This scenario covers workplace interventions, primary-care prevention, and the promotion of physical activity.

The *senior_functional_ageing* scenario targets the older population (471,412), with participation of 117,853 and effective participation of 88,389.75, corresponding to a share of $Share = 0.0415$. The functional effect is $FC = 0.026$, the lowest value among the scenarios, since this population has limited potential for improvement. The annual cost amounts to 21,999,420 EUR. This scenario is directed primarily at maintaining functional capacity and preventing falls and premature functional decline.

The *integrated_life_course* scenario combines all three life stages and covers the entire population (2,130,462), with participation of 958,707.9 and effective participation of 623,160.135, corresponding to a share of $Share = 0.2925$. The functional effect is $FC = 0.11$, the highest value among all scenarios, since it includes cumulative effects across the life course. The annual cost amounts to 113,077,250.5 EUR. This scenario represents a combination of measures in schools, in the workplace, and among the older population, enabling synergistic effects.

The timing of scenario implementation is determined deterministically by a linear rollout function

$$rollout_t = \frac{t - 2026}{2050 - 2026},$$

which gives a value of 0 for 2026 and 1 for 2050. The effective functional effect over time is therefore defined as

$$FC_{s,t} = FC_s \cdot rollout_t.$$

Scenario costs are treated in the model as exogenously determined annual expenditures $Cost_s$, which do not change over time and are not endogenously dependent on the scale of effects achieved. These costs enter directly into the calculation of scenario net effects and serve as the basis for comparing cost-effectiveness.

The entire scenario construction is therefore fully determined by a set of parameters, either estimated from regression models or set as calibrated or literature-based parameters, with each scenario affecting the outcome variables exclusively through a chain of clearly defined equations. This specification allows for complete traceability of results, since every effect can be directly explained as a function of the input parameters, the share of the population included, and the timing dynamics of implementation.

6. Results

The simulation results show that scenarios differ significantly in the relationship between the scale of population reach, health effects, cost-effectiveness, and broader socioeconomic benefits. The strongest absolute effect on healthy life years is achieved by the *integrated_life_course* scenario, which in the deterministic simulation generates 404,717 discounted additional healthy life years. It is followed by the *active_mobility* scenario with 142,209 additional HLYs, then *school_push* with 74,993 HLYs, *adult_prevention* with 67,999 HLYs, and *senior_functional_ageing* with 15,735 HLYs. The difference between scenarios is therefore pronounced as the integrated life-course approach achieves not only the largest effect but also the largest absolute social net benefit.

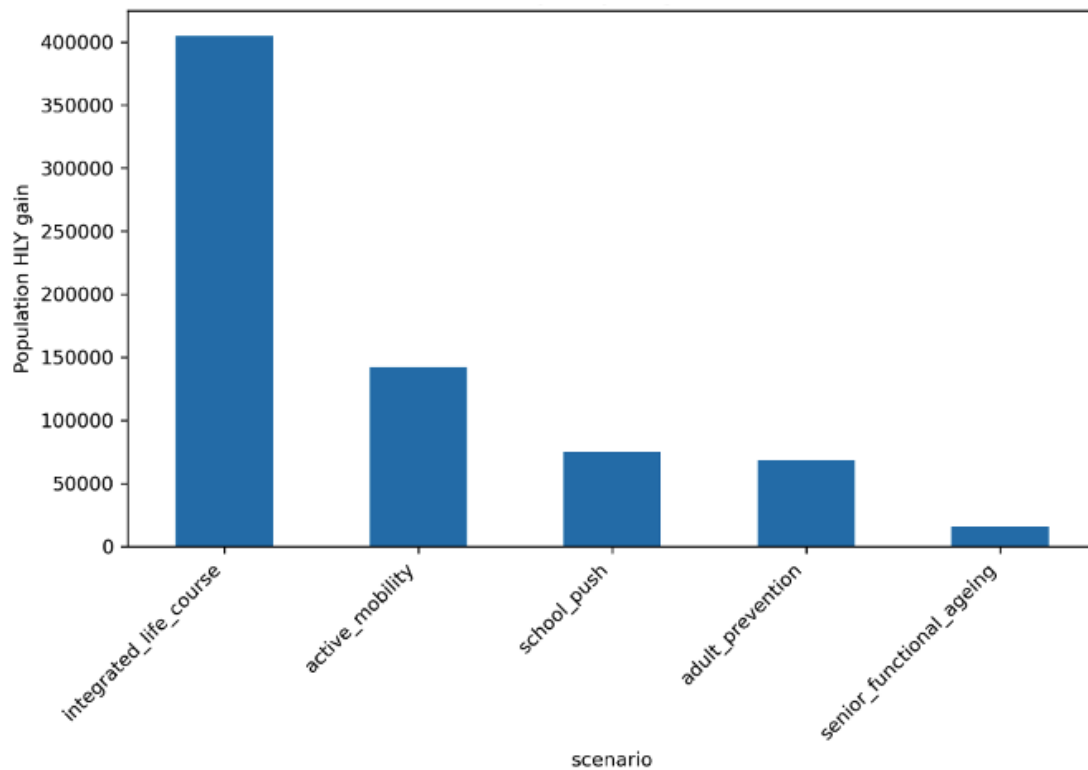


Figure 4 : Discounted Healthy Life Year Gain versus Baseline

From a cost-effectiveness perspective, the *active_mobility* scenario seems the most efficient, with a baseline ICER of 3,662 EUR per additional healthy life year, or 2,726 EUR when broader benefits are taken into account. The *school_push* scenario has an ICER of 4,877 EUR, and *integrated_life_course* has 4,890 EUR. This means the integrated scenario is the strongest in absolute effect, while *active_mobility* is the most favorable in terms of cost per healthy life year gained. The *adult_prevention* and *senior_functional_ageing* scenarios are less efficient, as shown ICER of 14,779 EUR, and the latter 24,707 EUR per HLY respectively.

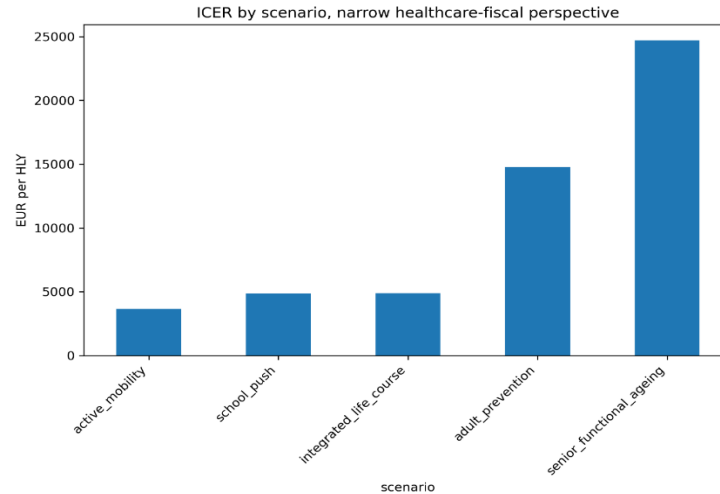


Figure 5 : ICER by Scenario, Narrow Healthcare-Fiscal Perspective

Since the analysis also includes broader benefits, the extended perspective is more important for understanding the economic rationale of the measures. When broader benefits are taken into account, the ICER decreases for all scenarios. *Active_mobility* remains the most favorable, with an extended ICER of 2,726 EUR per HLY, followed by *integrated_life_course* at 3,923 EUR and *school_push* at 4,138 EUR. This confirms that scenarios with large population reach and relatively low costs have the most favorable benefit-cost ratio.

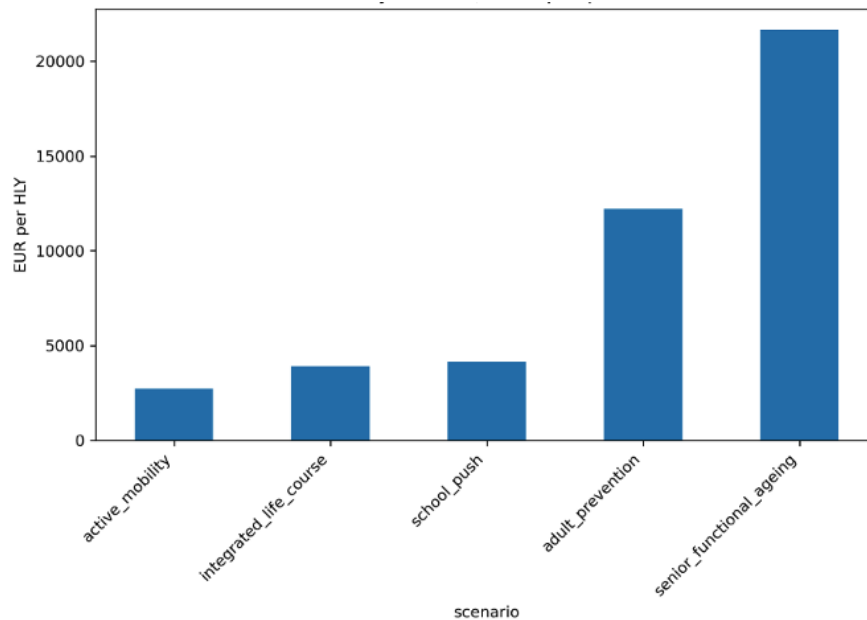


Figure 6 : ICER by scenario, Broad Perspective

When evaluated from a societal perspective, the results are even more pronounced. At a willingness to pay of 20,000 EUR per additional healthy life year, *integrated_life_course* achieves the highest social net benefit, at 6.12 billion EUR. *Active_mobility* generates 2.32 billion EUR, *school_push* 1.13 billion EUR, *adult_prevention* 355 million EUR, while *senior_functional_ageing* is still negative at this threshold, at -74 million EUR. This shows that the integrated scenario is the most compelling as a systemic public-policy choice if the goal is to maximize total social surplus.

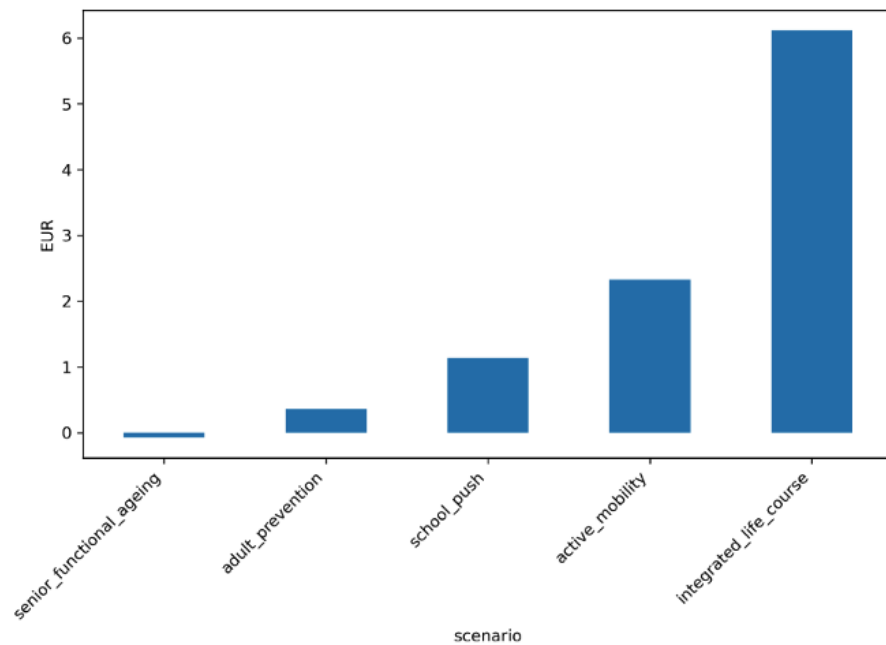


Figure 7 : Social net benefit at 20,000 EUR per HLY

Additional insight is provided by the benefit-cost ratio at 20,000 EUR per HLY. The highest benefit-cost ratio is achieved by *active_mobility*, at 5.62. It is followed by *integrated_life_course* at 4.21 and *school_push* at 4.20. *Adult_prevention* remains positive at 1.52, while *senior_functional_ageing* is below the threshold of one, at 0.93. This implies that the first three scenarios are economically very well justified, *adult_prevention* is worthwhile but less efficient, and *senior_functional_ageing* requires more cautious interpretation, since its effects are not large enough relative to its costs, except at higher willingness-to-pay values.

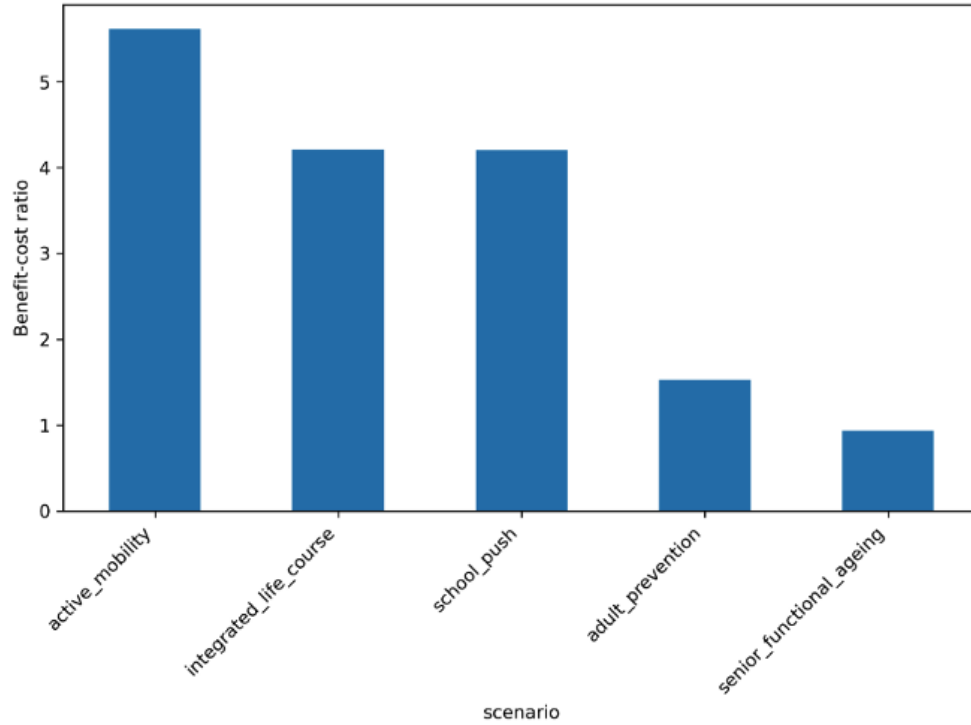


Figure 8 : Benefit-cost ratio at 20,000 EUR per HLY

The Monte Carlo analysis confirms the robustness of the main results. Across all scenarios, the probability of a positive HLY effect is high, roughly between 86% and 87%. However, scenarios differ in the dispersion of effects. The largest expected HLY benefit is found for *integrated_life_course*, with a median of 279,941 HLYs and a 95th percentile of 584,565 HLYs. *Active_mobility* has a median of 98,747 HLYs, *school_push* 52,043 HLYs, *adult_prevention* 47,567 HLYs, and *senior_functional_ageing* 10,946 HLYs. This confirms that the integrated scenario is not only the strongest deterministically, but also remains the most dominant under parameter uncertainty.

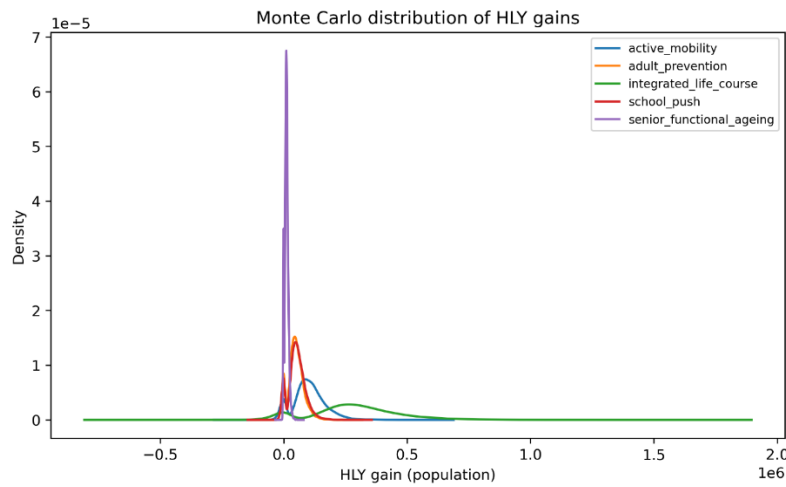


Figure 9 : Monte Carlo distribution of HLY gains

At the cost-effectiveness level, the Monte Carlo analysis shows that median ICERs remain low for the three key scenarios. The median ICER is 4,330 EUR for active_mobility, 5,560 EUR for integrated_life_course, and 5,809 EUR for school_push. For adult_prevention the median is already substantially higher, at 16,540 EUR, and for senior_functional_ageing it is 28,933 EUR. This means that the ranking of scenarios does not change substantially under uncertainty analysis: the most robust are the integrated scenario, active mobility, and the school scenario.

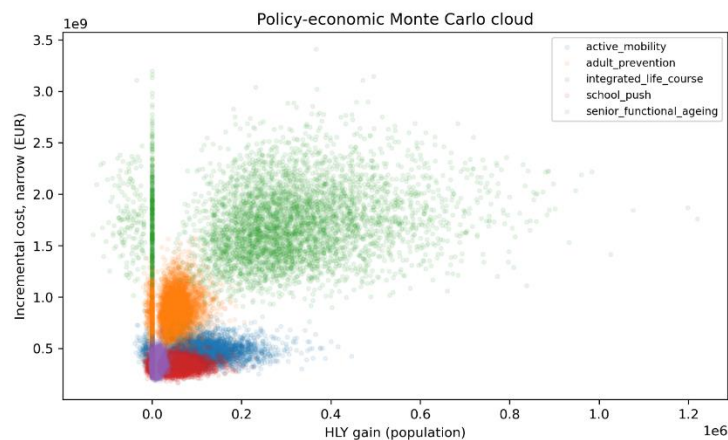


Figure 10 : Policy – Economics Monte Carlo cloud

Particularly important is the split of effects between working-age and non-working years. *Active_mobility* generates 99,547 HLYs in the working-age population and 42,663 HLYs outside it. *Integrated_life_course* generates 242,830 working-age HLYs and 161,887 non-working-age HLYs. For *school_push*, the ratio is different: 26,247 HLYs in the working-age component and 48,745 non-working-age HLYs, reflecting the long-term nature of the effects of school interventions. *Adult_prevention* has the largest share of benefits in the working-age population, while *senior_functional_ageing*, as expected, is directed primarily at the non-working age period.

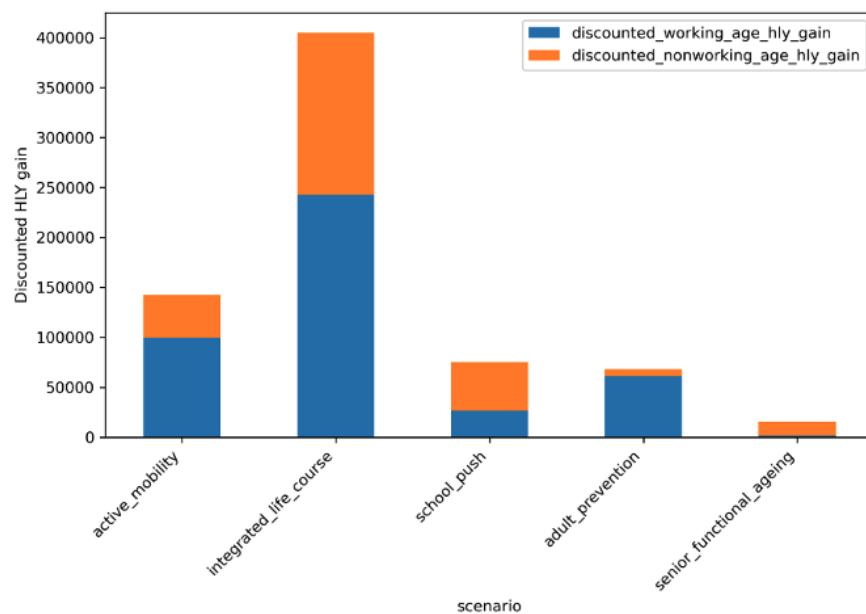


Figure 11 : Discounted HLY gains – working age versus Non working age

When additional effects on productivity, absenteeism, early retirement, and fiscal feedback effects are included in the model, the basic conclusions do not change but are further reinforced. In the extended labour-augmented analysis, *active_mobility* has the lowest ICER, 2,717 EUR per HLY, followed by *integrated_life_course* at 3,915 EUR, then *school_push* at 4,134 EUR. *Adult_prevention* remains at 12,208 EUR, and *senior_functional_ageing* at 21,683 EUR. The additional labor-market effects are numerically smaller than the health and HLY effects, but they improve the fiscal picture, especially for scenarios with large working-age reach.

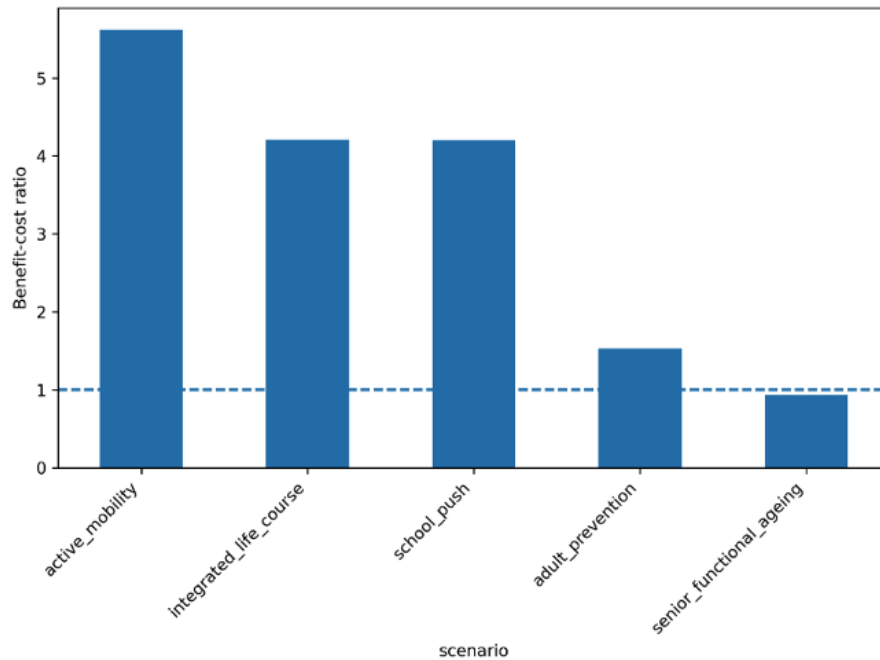


Figure 12 : Labor-augmented benefit-cost ratio at 20,000 EUR per HLY

The final decision from an economic-policy perspective is therefore relatively clear. If the goal is to maximize the total health effect and social net benefit, the best choice is *integrated_life_course*. If the goal is the lowest cost per healthy life year gained, *active_mobility* is the most favorable. If the goal is a long-term preventive policy with relatively low cost and good cost-effectiveness, *school_push* is the most desirable option. *Adult_prevention* is economically acceptable primarily at higher willingness-to-pay thresholds, while *senior_functional_ageing* is weaker on its own but can remain worthwhile as part of an *integrated_life-course* approach.

47_policy_decision_matrix_labour.png

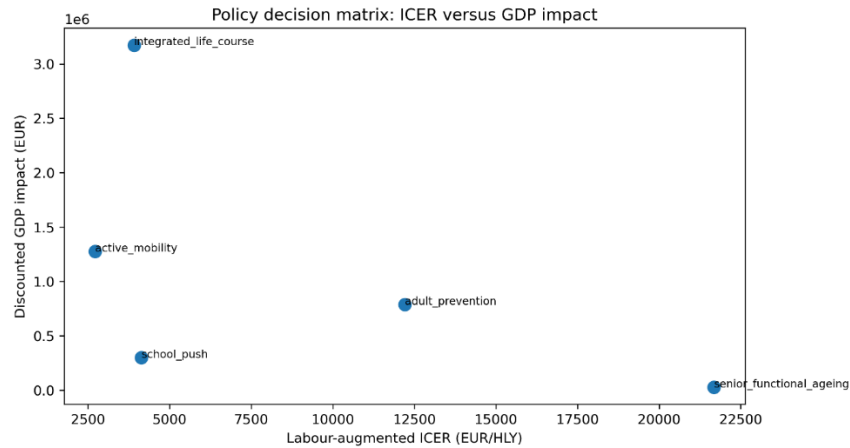


Figure 13 : Policy Decision Matrix : ICER versus GDP impact

7. Conclusion

This paper develops an integrated simulation framework linking physical fitness to population health, healthcare expenditure, labor-market outcomes, and broader economic consequences. Using functional capacity as the central transmission mechanism, the model moves beyond approaches that treat physical activity exclusively as a behavioral input. Rather, it represents physical fitness as an accumulated physiological resource through which preventive interventions may significantly influence obesity, healthy life years, healthcare costs, labor market outcomes.

The results highlight that the relative performance of policy scenarios depends on the objective being considered. The integrated life-course scenario produces the largest overall gains in healthy life years and the greatest social net benefit, reflecting its broad population reach and its cumulative effects across different stages of life. Active mobility scenario generates the lowest cost per additional healthy life year and the highest benefit-cost ratio, making it the most economically efficient individual scenario. The school-based scenario also performs favorably, particularly as a long-term preventive investment. Adult prevention produces positive benefits

but is less cost-effective than the broader population-based approaches, while the senior functional aging scenario generates comparatively smaller returns when implemented independently.

The uncertainty analysis generally supports these findings. Although the magnitude of the estimated effects varies across simulations, the integrated life-course, active-mobility, and school-based scenarios remain the strongest alternatives under most assumptions. The inclusion of productivity, absenteeism, early retirement, and fiscal feedback modestly improves their economic performance but does not substantially alter the overall ranking. The principal economic value of the interventions therefore arises from improvements in health and healthy life years, with labor market effects providing an additional source of benefit.

From a policy perspective, the findings support a coordinated and gradually implemented strategy rather than reliance on a single isolated intervention. Active-mobility and school-based measures appear to provide feasible starting points as they combine broad reach, favorable cost-effectiveness, and established institutional delivery mechanisms. Workplace and primary care prevention could subsequently extend the strategy to working age adults, while strength, balance, mobility, and fall prevention programs should remain a crucial component of healthy aging policy. Integrating these measures across education, health, labor, transport, and municipal policy may generate greater value than implementing them independently.

Nevertheless, these findings should be interpreted within the structure and assumptions of the model. Several parameters are based on calibrated relationships or conservative priors as consistent longitudinal information is not available for every transmission pathway. Thus, the estimates should be interpreted as scenario-based projections rather than causal forecasts of the exact effects that would follow from policy implementation. Future research may validate the functional capacity pathways using individual level longitudinal data, refine the estimates of intervention costs and adherence, and assess how effects differ across socioeconomic and demographic groups.

Taken together, the findings emphasize the value of treating physical capacity as a measurable form of population capital. The policies that improve functional capacity across the

life course may significantly contribute not only to better health but also to greater economic and fiscal sustainability. More broadly, the model provides a transparent platform through which alternative preventive policies can be compared using a common set of health, economic, and social outcomes.

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